

Assignment – 5

Question – 1:

Code:

```
1 // Write a C program to print all the natural numbers in reverse using while loop.
2 #include <stdio.h>
3 int main() {
4     int n;
5     printf("Enter the value of N: ");
6     scanf("%d", &n);
7     for(int i = n; i > 0; i--) printf("%d ", i);
8     printf("\n");
9 }
```

Output:

```
students@cseblabadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_5$ ./q1
Enter the value of N: 15
15 14 13 12 11 10 9 8 7 6 5 4 3 2 1
```

Question – 2:

Code:

```
1 // Print all the even numbers between 1 to 100 using while loop.
2 #include <stdio.h>
3 int main() {
4     for(int i=0; i<101; i++) {
5         if(i%2 == 0) printf("%d ", i);
6     }
7     printf("\n");
8 }
```

Output:

```
students@cseblabadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_5$ ./q2
0 2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50 52 54 56 58 60 62 64 66 68 70 72 74 76 78 80 82 84 86 88 90 92 94 96 98 100
```

Question – 3:

Code:

```
1 // Write a C program to count the number of digits in a number.
2 #include <stdio.h>
3 int main() {
4     int n, _n, digits = 0;
5     printf("Enter a number: ");
6     scanf("%d", &n);
7     _n = n;
8     while(n!=0) {
9         n/=10;
10        digits++;
11    }
12    printf("The number of digits in %d is %d.\n", _n, digits);
13 }
```

Output:

```
students@cseblabadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_5$ ./q3
Enter a number: 45678
The number of digits in 45678 is 5.
```

Question – 4:

Code:

```
1 // Write a C program to find LCM of 2 given numbers.
2 #include <stdio.h>
3 int main() {
4     int a, b, hcf=1, lcm;
5     printf("Enter two numbers: ");
6     scanf("%d %d", &a, &b);
7     int min = (a<b) ? a : b;
8     for(int i=1; i<=min; i++) { if(a%i==0 && b%i==0) hcf = i; }
9     lcm = a*b/hcf;
10    printf("The LCM of %d and %d is %d\n", a, b, lcm);
11 }
```

Output:

```
students@cse1abiadmn-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_5$ ./q4
Enter two numbers: 3 5
The LCM of 3 and 5 is 15
```

Question – 5a:

Code:

```
1 /*
2 Pattern:
3 *
4 * *
5 * * *
6 * * * *
7 * * * * *
8 */
9 #include <stdio.h>
10 int main() {
11     int n=5;
12     for(int i=0; i<n; i++) {
13         for(int j=n-i; j>0; j--) printf(" ");
14         printf("*");
15         if(i!=0) {
16             for(int k=2*i-1; k>0; k--) printf(" ");
17             printf("*");
18         }
19         printf("\n");
20     }
21     for(int i=0; i<2*n+1; i++) printf("*");
22     printf("\n");
23 }
```

Output:

```
students@cse1abiadmn-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_5$ ./q5a
*
* *
* * *
* * * *
* * * * *
*****
```

Question – 5b:

Code:

```
1/*
2Pattern:
31
40 1
50 1 0
61 0 1 0
71 0 1 0 1
8*/
9#include <stdio.h>
10int main() {
11    int n = 5, a=1;
12    for(int i=0; i<n+1; i++) {
13        for(int j=0; j<i+1; j++) {
14            if(a==1) { printf("%d ", a); a = 0; }
15            else if(a==0) { printf("%d ", a); a = 1; }
16        }
17        printf("\n");
18    }
19}
```

Output:

```
students@cse1abiadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_5$ ./q5b
1
0 1
0 1 0
1 0 1 0
1 0 1 0 1
0 1 0 1 0 1
```

Question – 5c:

Code:

```
1/*
2Pattern:
31      1
42      2
53      3
64      4
75
84      4
93      3
102      2
111      1
12*/
13#include <stdio.h>
14int main() {
15    int n=5;
16    for(int i=1; i<n; i++) {
17        for(int j=i; j>0; j--) printf(" ");
18        printf("%d", i);
19        for(int k=2*(n-i)-1; k>0; k--) printf(" ");
20        printf("%d", i);
21        printf("\n");
22    }
23    for(int i=n; i>0; i--) {
24        for(int j=i; j>0; j--) printf(" ");
25        printf("%d", i);
26        if(i!=n) {
27            for(int k=2*(n-i)-1; k>0; k--) printf(" ");
28            printf("%d", i);
29        }
30        printf("\n");
31    }
32}
```

Output:

```
students@cse1abiadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_5$ ./q5c
1      1
2      2
3      3
4      4
5
6      4
7      3
8      2
9      1
```

Question – 6:

Code:

```
1 // Write a C Program to calculate product of digits of a number.
2 #include <stdio.h>
3 int main() {
4     int n, _n, prod = 1;
5     printf("Enter a number: ");
6     scanf("%d", &n);
7     _n = n;
8     while(n!=0) {
9         prod *= n%10;
10        n/=10;
11    }
12    printf("The product of digits of %d is %d.\n", _n, prod);
13 }
```

Output:

```
students@cse1abadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_5$ ./q6
Enter a number: 345
The product of digits of 345 is 60.
```